

BOARD OF INTERMEDIATE EDUCATION KARACHI · BIEK

Computer Science II

Lecture Notes

Class XII · Science General & Humanities · Printed lectures for college and self-study

Sindh Curriculum 2019

BIEK Model Paper 2026

English + Urdu

C / VB / MS Access

What is in this book

Paper	Computer Science Paper II · 75 marks · choose ONE option
Option I	Programming using C + Database / MS Access
Option II	Programming using Visual Basic + Database / MS Access
Also covered	Pointers, OOP (C++), file handling, SDLC, multimedia, wireless — Sindh Grade XII units

Unofficial teaching notes for students of BIEK-affiliated colleges. Definitions, programs and diagrams follow the Sindh HSC Computer Science curriculum and the official BIEK paper pattern (Section A MCQs, Section B short, Section C detailed). Turbo C / void main() style is shown because that is what Karachi board answer books still expect.

Lecture index

Unit 0: Paper II map

1. BIEK Paper II: Option I (C) or Option II (Visual Basic)

Unit C1: Programming using C

2. C language, structure of a program and Turbo C IDE ★

Unit C2: Programming using C

3. Data types, variables, constants and input/output ★

Unit C3: Programming using C

4. Operators, precedence and expressions ★

Unit C4: Programming using C

5. Selection: if, if-else, nested if and switch ★

Unit C5: Programming using C

6. Loops: for, while, do-while, break, continue, patterns ★

Unit C6: Programming using C

7. Arrays and strings ★

Unit C7: Programming using C

8. Functions: prototype, call, return, types ★

Unit 2: Pointers (Sindh XII + C paper)

9. Pointers and the address operator ★

Unit 3–4: OOP using C++ and file handling

10. OOP in C++: class, object, constructor, inheritance, polymorphism ★

Unit VB1: Programming using Visual Basic

11. Visual Basic IDE, forms and ActiveX controls ★

Unit VB2: Programming using Visual Basic

12. VB language: operators, if, Select Case, loops, arrays, functions ★

Unit 5: Database fundamentals

13. DBMS, models, ER diagrams, keys and normalisation ★
14. MS Access objects, data types and SQL ★

Unit 1: System Development Life Cycle

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18. Paper II Option II mock (VB + Access) with answers ★

BIEK Paper II: Option I (C) or Option II (Visual Basic)

One college period · 40–45 minutes · Paper II map

What you will learn

- Choose the option you studied in college — you cannot mix C and VB in one paper
- State that Database / MS Access is common to both options
- Know Section A 15 MCQs, Section B 10 of 15 shorts, Section C 3 of 5 longs (2026 model)

LEARN IT

Paper II is 75 marks. The front page says: attempt ONE option — Programming using C OR Programming using Visual Basic. Database questions (keys, MS Access objects, data models, SQL) appear in BOTH options.

	Option I — C	Option II — Visual Basic
Language	Turbo C / ANSI C in answer book	VB 6 style forms + code
Typical programs	factorial, table, greater of two, patterns, functions	even numbers, sum/average, tax slabs, calculator form
Theory	tokens, operators, loops, arrays, functions, pointers	controls, DIM, loops, arrays, built-in functions, errors
Shared	DBMS, tables, keys, ER, Access data types, SQL	same

Sindh Curriculum 2019 Grade XII also includes SDLC, pointers, OOP with C++, file handling, multimedia and wireless. Colleges that follow the new book still need C or VB for the BIEK paper, so this pack teaches both the board options and the extra Sindh units.

HOW BIEK ASKS THIS

2026 Option I shorts: structure of C, #include/#define, multiplication table, break vs continue, reserved words, address operator, factorial, scanf vs getchar, primary vs foreign key, expression values, DBMS components, function to print greater number, maths→C, nested-loop patterns, table/query/report.

COMMON BOARD MISTAKE

void is a reserved word, so it is an invalid identifier. Identifiers cannot start with a digit or contain a space: 1st number is invalid; std_marks is valid.

اردو میں سمجھیں

بارہویں کا پیپر ٹو دو آپشن میں ہے: C زبان یا Visual Basic. دونوں میں MS Access / ڈیٹا بیس آتا ہے۔ جو آپشن کالج میں پڑھا ہے وہی حل کریں۔

CHECK YOURSELF

1. Can you answer a C long question in the VB option?
2. Is MS Access only in Option II?

Answers: 1. No — pick one option for the whole paper · 2. No — database is in both options

MCQs

1. Every C program must contain:
 - a. start()
 - b. getch()
 - c. main()
 - d. printf()
2. Dennis Ritchie founded:
 - a. C
 - b. BASIC
 - c. JAVA
 - d. PASCAL
3. In MS Access a row is also called a:
 - a. Field
 - b. Column
 - c. Record
 - d. Query

Key: 1=c, 2=a, 3=c

C language, structure of a program and Turbo C IDE ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write the basic structure of a C program
- Explain #include and #define with examples
- Use Turbo C shortcuts and know compiler vs editor vs debugger
- List reserved words and rules of identifiers

C was designed by Dennis Ritchie at Bell Labs (1972) to write UNIX. It is a middle-level, compiled, case-sensitive language. BIEK still expects Turbo C style in many colleges: #include <stdio.h>, #include <conio.h>, void main(), clrscr(), getch().

```
#include <stdio.h>
#include <conio.h>
#define PI 3.14
void main()
{
    clrscr();
    printf("Board of Intermediate Education Karachi\n");
    getch();
}
```

Part	Job
#include <stdio.h>	Preprocessor: copies standard I/O functions (printf, scanf)
#include <conio.h>	Turbo C console: clrscr, getch
#define PI 3.14	Manifest constant — name replaced before compile
void main()	Execution starts here (ISO C prefers int main(void))
Declarations	Variables before use in C89 / Turbo C
Statements	End with semicolon
getch()	Wait for a key so the output screen stays

Identifiers and keywords

- Letters, digits, underscore; cannot start with a digit; no spaces or special symbols; cannot be a keyword.
- Valid: std_marks, NUMBER5, Number_5. Invalid: 5number, 1st number, void, int, float.
- Keywords (reserved): auto, break, case, char, const, continue, default, do, double, else, enum, extern, float, for, goto, if, int, long, register, return, short, signed, sizeof, static, struct, switch, typedef, union, unsigned, void, volatile, while.

Escape sequence. A backslash code inside a string: \n new line, \t tab, \\ backslash, \a beep, \" quote, \0 string terminator.

HOW BIEK ASKS THIS

2025/26 shorts: structure of C; #include or #define; reserved words; valid/invalid identifiers; why C is case sensitive; IDE shortcuts.

اردو میں سمجھیں

C پروگرام #include سے شروع ہوتا ہے، main() سے چلتا ہے۔ #define نام کو قدر سے بدل دیتا ہے۔ (int, Reserved words void, for نام نہیں بن سکتے۔)

CHECK YOURSELF

1. Who developed C?

2. What does \n do?

Answers: 1. Dennis Ritchie · 2. Moves the cursor to a new line

MCQs

1. C programs are converted to machine language by a:
 - a. Editor
 - b. Assembler
 - c. Compiler
 - d. Debugger
2. This is not a valid identifier:
 - a. Number
 - b. 5number
 - c. NUMBER5
 - d. Number_5
3. Comments in C are enclosed in:
 - a. /* ... */
 - b. /* ... /*
 - c. */ ... /*
 - d. ** **
4. Names of variables and functions are called:
 - a. header files
 - b. identifiers
 - c. loops
 - d. structures

Key: 1=c, 2=b, 3=a, 4=b

Short questions (3 marks each)

1. Write the basic structure of a C program.
2. Define #include or #define with two examples.
3. What is an IDE? Write a few shortcut keys.
4. Identify valid/invalid: void, std_marks, 1st number.
5. Why is C called case-sensitive? Define three reserved words.

Model short answers

1. Preprocessor directives, global declarations, main(), local declarations, statements, closing brace.
Show a 8-line skeleton as above.
2. #include <stdio.h> brings printf. #include <math.h> brings sqrt. #define PI 3.14 and #define MAX 100 create constants.
3. IDE = editor+compiler+debugger. Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 user screen, F8 step.
4. void invalid (keyword). std_marks valid. 1st number invalid (digit start + space).
5. TOTAL and total are different. Keywords: int, return, while.

Long questions (board Section C)

1. Explain the structure of a C program with a labelled example. Discuss tokens: keywords, identifiers, constants, operators.

Data types, variables, constants and input/output ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- List C data types and format specifiers
- Use printf, scanf, getchar, getch, puts, gets (exam style)
- Convert algebraic expressions into C

Type	Typical size (Turbo C 16-bit)	Specifier	Example
char	1 byte	%c	grade = 'A'
int	2 bytes	%d	n = 25
float	4 bytes	%f	avg = 72.5
double	8 bytes	%lf	pi = 3.14159
long	4 bytes	%ld	pop = 16000000L

Constants: 25 integer, 3.14 float, 'A' character, "BIEK" string. Variables must be declared before use: int roll; float marks;

```
int a, b, sum;
printf("Enter two numbers: ");
scanf("%d%d", &a, &b); /* & is the address operator */
sum = a + b;
printf("Sum = %d\n", sum);
```

Function	Use
printf	Formatted output to screen
scanf	Formatted input; needs & for scalars
getchar	Read one character (waits for Enter)
getch	Read one character without echo (conio.h)
gets / puts	String line I/O (unsafe in modern C, still in old papers)

Algebra → C (every year)

Mathematics	C expression
$A = \frac{1}{2} \times \text{base} \times \text{height}$	<code>A = 0.5 * base * height;</code>
$x = 3ab^3 + 3a^2b$	<code>x = 3*a*b*b*b + 3*a*a*b;</code>
$d = b^2 - 4ac $	<code>d = fabs(b*b - 4*a*c); /* math.h */</code>
$A = \pi r^2$	<code>A = 3.14 * r * r;</code>
$(a+b)^2 / ab$	<code>((a+b)*(a+b))/(a*b)</code>
$y = (a+b)^n$	<code>y = pow(a+b, n);</code>

WORKED EXAMPLE / BOARD STYLE

If $a=2$, $b=5$, $c=10$:

$200*((a*10)+c)$ wait — follow the paper's parentheses exactly.

$a*b*10/c+a+c = 2*5*10/10+2+10 = 100/10+2+10 = 10+2+10 = 22$ (integer division if all int).

$a-(c/b)/2+100 = 2-(10/5)/2+100 = 2-2/2+100 = 2-1+100 = 101$ with int division of $10/5$ first.

COMMON BOARD MISTAKE

`scanf("%d", &n)` — forgetting `&` is the #1 runtime bug. `%` with float is undefined; `%` is for integers only.

اردو میں سمجھیں

int عدد صحیح، float اعشاری، char ایک حرف، `%%`، `%f`، `%c`، `%d` فارمیٹ۔ `scanf` میں `&` پتہ دیتا ہے۔
ریاضی کو C میں `*` / اور `pow()` سے لکھیں۔ $\frac{1}{2}$ کو `*0.5` بنائیں۔

CHECK YOURSELF

1. Format specifier for int?

2. Why `&` in `scanf`?

Answers: 1. `%d` · 2. Pass the address so the value can be stored in the variable

MCQs

1. `%` operator is only used with:

- a. float
- b. int
- c. double
- d. string

2. The address operator is:

- a. `*`
- b. `&`
- c. `%`
- d. `#`

Key: 1=b, 2=b

Short questions (3 marks each)

1. How does `scanf()` differ from `getchar()`?
2. What is the address operator? Explain with example.
3. Convert $A = \frac{1}{2} \text{ base} \cdot \text{height}$ and $A = \pi r^2$ into C.
4. Define increment and decrement operators.

Model short answers

1. scanf reads formatted data (numbers, strings) using specifiers; getchar reads a single character from the keyboard buffer.
2. & gives the memory address. `int x=5; printf("%u", &x);` prints the address of x. `scanf("%d", &x)` stores input at that address.
3. `A=0.5*base*height;` `A=3.14*r*r;` (or `PI*r*r`).
4. ++ adds 1, -- subtracts 1. Pre (++k) changes then uses; post (k++) uses then changes.

Long questions (board Section C)

1. What are different types of operators in C? Explain with examples. Also discuss printf and scanf.

Operators, precedence and expressions ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Classify arithmetic, relational, logical, assignment, increment, conditional, bitwise operators
- Apply precedence: () then ++/-- then * / % then + -
- Trace small programs with x++ vs ++x

Class	Operators	Result
Arithmetic	+ - * / %	% remainder, integers only
Relational	> < >= <= == !=	1 (true) or 0 (false)
Logical	&& !	compound conditions
Assignment	= += -= *= /= %=	x += 2 means x = x+2
Increment	++ --	pre or post
Conditional	? :	x = (a>b)? a : b;
Bitwise	& ^ ~ << >>	bit level (rare in Paper II but listed in books)

LEARN IT

Precedence (high → low): () ++ -- * / % + - relational logical assignment. Same-level * / % go left to right.

Integer division: 5/2 is 2, not 2.5. Use 5/2.0 or a float variable.

WORKED EXAMPLE / BOARD STYLE

If x = 1:

printf("x=%d\n", x); → x=1

printf("x=%d\n", x + x); → x=2

printf("x=%d\n", x); → x=1 (x was never assigned a new value)

Post-increment example: k++ is post; ++k is pre. MCQ: k++ is post increment.

```
int a = 5, b = 2, c;
c = a / b;          /* 2 */
c = a % b;          /* 1 */
c = (a > b) && (b > 0); /* 1 */
c = (a > b) ? a : b;  /* 5 */
```

HOW BIEK ASKS THIS

Long 2026 Q4: types of operators in C. Make a table with one example each. Trace questions: never change a variable unless you see =, ++ or --.

CHECK YOURSELF

1. Value of 17 % 5?
2. Is k++ pre or post increment?

Answers: 1. 2 · 2. Post

MCQs

1. An example of post increment is:
 - a. k++
 - b. ++k
 - c. k+=
 - d. k=+
2. Highest precedence among + - / % is:
 - a. +
 - b. -
 - c. /
 - d. % — actually / and % are equal and higher than + -
3. == is a:
 - a. assignment
 - b. relational operator
 - c. loop
 - d. header

Key: 1=a, 2=c, 3=b

Short questions (3 marks each)

1. Define increment and decrement with examples.
2. If x=1, output of three printf lines using x, x+x, x.
3. Write C expressions for $(x^2 - y^2)(x+y)/(a+b)$.

Model short answers

1. ++k pre-increment, k++ post. -- same for decrement. int k=5; printf("%d", k++); prints 5 then k becomes 6.
2. 1 then 2 then 1.
3. $((x*x - y*y)*(x+y))/(a+b)$

Long questions (board Section C)

1. What are different types of operators used in a C program? Explain with examples and precedence.

Selection: if, if-else, nested if and switch ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write if, if-else, else-if ladder and nested if
- Write switch-case with break and default
- Draw the switch logical diagram
- Program: greater of two, pass/fail, leap year

```
if (marks ≥ 40)
    printf("Pass");
else
    printf("Fail");
```

```
if (n > 0) printf("Positive");
else if (n < 0) printf("Negative");
else printf("Zero");
```

switch (board long question)

```
int ch;
printf("1.Add 2.Sub 3.Mul\n");
scanf("%d", &ch);
switch (ch) {
    case 1: printf("Add"); break;
    case 2: printf("Sub"); break;
    case 3: printf("Mul"); break;
    default: printf("Invalid");
}
```

LEARN IT

switch works on int or char (not float). Each case should end with break, otherwise execution falls through. default is optional but expected in board answers.

Evaluate expression → Match case → Execute statements → break → leave switch

WORKED EXAMPLE / BOARD STYLE

Greater of two using a function (2026 Q2 xii):

```
int greater(int x, int y) {
    if (x > y) return x;
    else return y;
}
void main() {
    int a, b;
    scanf("%d%d", &a, &b);
    printf("Greater = %d", greater(a, b));
}
```

HOW BIEK ASKS THIS

2026 Section C Q3: Define switch with logical diagram. Draw a diamond or a multi-way box: expression, case 1, case 2, default.

اردو میں سمجھیں

if شرط سچی ہو تو بلاک چلتا ہے۔ switch ایک متغیر کی کئی قیمتوں پر چلتا ہے، ہر case کے بعد break لکھیں۔

CHECK YOURSELF

1. Can switch test a float?
2. What if you omit break?

Answers: 1. No · 2. Fall-through: the next case also runs

MCQs

1. switch is a:
 - a. loop
 - b. selection / decision structure
 - c. function
 - d. header
2. default in switch runs when:
 - a. always
 - b. no case matches
 - c. every case
 - d. at compile time

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define switch with a small example.
2. Write a program that prints the greater of two integers using a function.
3. Differentiate if-else and switch.

Model short answers

1. switch(expression) { case c1: ... break; default: ... } selects one case by matching integers/chars.
2. Full greater() program as in the lecture.
3. if-else handles ranges and floats; switch is multi-way on discrete int/char values and is cleaner for menus.

Long questions (board Section C)

1. Define switch statement with a logical diagram and a complete C program (menu or grade).

Loops: for, while, do-while, break, continue, patterns ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write for, while and do-while and compare them
- Use break and continue
- Print multiplication tables, factorial, nested-loop patterns

	for	while	do-while
Test	At the top	At the top	At the bottom — runs at least once
Best when	Count known	Condition, count unknown	Menus, "repeat until"
Header	for(s;c;u)	while(c)	do { } while(c);

```
/* table of n */
int n, i;
scanf("%d", &n);
for (i = 1; i ≤ 10; i++)
    printf("%d x %d = %d\n", n, i, n * i);
```

```
/* factorial : 4! = 24 */
int n, i; long f = 1;
scanf("%d", &n);
for (i = 1; i ≤ n; i++)
    f = f * i;
printf("%ld", f);
```

break vs continue

- break: leave the loop immediately.
- continue: skip the rest of this iteration and go to the next test / increment.

```
int i;
for (i = 1; i ≤ 5; i++) {
    if (i == 3) continue;
    printf("%d ", i);    /* 1 2 4 5 */
}
```

Nested loops — 2026 pattern

```
int i, j;
for (i = 1; i ≤ 5; i++) {
    for (j = 1; j ≤ i; j++)
        printf("%d", j);
    printf("\n");
}
/* 1
12
123
1234
12345 */
```

Number / Square: `for(i=1;i<=5;i++) printf("%d %d\n", i, i*i);`

HOW BIEK ASKS THIS

Programs to memorise: table, factorial, even 0–50, sum of n numbers, reverse digits, prime check, Fibonacci, patterns.

COMMON BOARD MISTAKE

do-while ends with a semicolon after `while(condition);` forgetting it is a syntax error. Infinite loop: `for(;;)` or `while(1)`.

اردو میں سمجھیں

for گنتی معلوم ہو تو، while شرط پر، do-while کم از کم ایک بار۔ break لوپ توڑتا ہے، continue اس چکر کو چھوڑتا ہے۔
ٹیبل، فیکٹوریل اور پیٹرن ہر سال آتے ہیں — زبانی لکھ کر یاد کریں۔

CHECK YOURSELF

1. $4! = ?$
2. Does do-while always run once?

Answers: 1. 24 · 2. Yes

MCQs

1. `for()` is used when we need:
 - a. Sequential structure
 - b. Selection structure
 - c. Iteration structure
 - d. Top down structure
2. This statement skips the rest of the loop body and continues from the top:
 - a. `break`
 - b. `resume`
 - c. `continue`
 - d. `skip`

Key: 1=c, 2=c

Short questions (3 marks each)

1. Write a program that prints the multiplication table of an inputted number.
2. Write a program that prints factorial of n (hint $4!=24$).
3. Differentiate while and do-while.
4. Define break and continue with an example.
5. Write a nested-loop program for the triangle 1 / 12 / 123 / ...

Model short answers

1. for i=1 to 10 print n x i = n*i.
2. f=1; for i=1..n f*=i; print f.
3. while tests first (may run 0 times). do-while tests after the body (at least once).
4. break exits; continue jumps to next iteration. Show a 4-line for-loop example.
5. Outer i=1..5, inner j=1..i print j, then newline.

Long questions (board Section C)

1. What is a loop? Explain for, while and do-while with programs. Discuss nested loops.

Arrays and strings ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Declare, initialise and traverse 1-D and 2-D arrays
- Explain string terminator '\0' and strlen, strcpy, strcat, strcmp
- Predict output of num[3] style MCQs

Array. A collection of elements of the same data type stored in contiguous memory, accessed by index starting at 0.

```
int num[] = {3, 5, 7, 10, 20};
printf("%d", num[3]);    /* 10 - index 3 is the fourth element */
```

```
int a[5], i, sum = 0;
for (i = 0; i < 5; i++) {
    scanf("%d", &a[i]);
    sum += a[i];
}
printf("Average = %.2f", sum / 5.0);
```

Two-dimensional array

```
int m[3][3], i, j;
for (i = 0; i < 3; i++)
    for (j = 0; j < 3; j++)
        scanf("%d", &m[i][j]);
```

Strings

A string is a char array ending with '\0'. char name[20] = "Ali"; /* 'A','l','i','\0' */

Function	Header	Use
strlen(s)	string.h	length without '\0'
strcpy(d,s)	string.h	copy s into d
strcat(d,s)	string.h	join s onto d
strcmp(a,b)	string.h	0 if equal, <0 or >0 if not

```
char s[30];
int i, n;
gets(s);          /* exam style */
n = strlen(s);
for (i = n - 1; i ≥ 0; i--)
    printf("%c", s[i]); /* reverse name */
```

HOW BIEK ASKS THIS

2025: What is an array? Declare and initialise. Output of num[3] with {3,5,7,10,20} is 10 — not 7. Students forget index 0.

COMMON BOARD MISTAKE

scanf("%s", name) — no & for arrays because the name is already an address. Overrunning the array size crashes the program.

اردو میں سمجھیں

Array ایک ہی قسم کی کئی قیمتیں، انڈیکس 0 سے۔ `num[3]` چوتھا عنصر ہے۔
سٹرنگ حروف کی صف ہے جو `\0` پر ختم ہوتی ہے۔ `strlen` لمبائی، `strcpy` نقل، `strcat` جوڑ، `strcmp` موازنہ۔

CHECK YOURSELF

1. Index of the first element?

2. `num[]={3,5,7,10,20}; num[3]?`

Answers: 1. 0 · 2. 10

MCQs

1. Array index in C starts from:

- a. 1
- b. 0
- c. -1
- d. 2

2. A string in C ends with:

- a. `\n`
- b. `\0`
- c. `\t`
- d. NULL pointer only

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is an array? Declare and initialise an integer array.
2. `int num[]={3,5,7,10,20};` what does `printf("%d", num[3])` print?
3. Write a program to reverse an inputted name using `strlen()`.
4. Name four string functions.

Model short answers

1. Same-type contiguous list. `int a[5]={1,2,3,4,5};` or `int a[]={10,20};`
2. 10.
3. Read string, loop from `strlen-1` down to 0 printing characters.
4. `strlen`, `strcpy`, `strcat`, `strcmp` (plus `strncpy` etc. if space).

Long questions (board Section C)

1. Explain array and its types with examples. Write a program to add two 3×3 matrices.

Functions: prototype, call, return, types ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- State advantages and characteristics of functions
- Differentiate built-in and user-defined; call by value vs reference
- Write programs that use functions (greater, factorial, area)

Function. A named, reusable block that performs a task. `main()` is itself a function. Advantages: modularity, reuse, easier testing, team work.

Prototype (declaration) → Definition (body) → Call (use)

```
int square(int n);           /* prototype */
void main() {
    printf("%d", square(5)); /* call – actual parameter 5 */
}
int square(int n) {          /* definition – n is formal parameter */
    return n * n;
}
```

Idea	Meaning
Built-in	<code>printf</code> , <code>scanf</code> , <code>sqrt</code> , <code>strlen</code> — already in libraries
User-defined	You write them
Return type	<code>void</code> if nothing is returned
Formal parameters	In the definition header
Actual parameters	In the call
Call by value	A copy is passed; original safe
Call by reference	Address (&) / pointer; function can change the original
Local vs global	Local inside block; global outside functions
Recursion	Function calls itself (factorial, Fibonacci)

HOW BIEK ASKS THIS

2026 long Q5: main characteristics of a function with example. List: name, return type, parameters, independent compilation, reuse, call by value/reference.

اردو میں سمجھیں

فونکشن نام والا بلاک ہے جو بار بار کام کرتا ہے۔ Prototype اعلان، تعریف جسم، کال استعمال۔ `return` قدر بھیجتا ہے۔

CHECK YOURSELF

1. Does main have to exist?
2. sqrt comes from which header?

Answers: 1. Yes — it is the entry point · 2. math.h

MCQs

1. A subprogram that returns a value is a:
 - a. function
 - b. report
 - c. macro only
 - d. file
2. Passing a copy of a variable is:
 - a. pass by value
 - b. pass by address
 - c. pass by reference
 - d. pass by pointer

Key: 1=a, 2=a

Short questions (3 marks each)

1. Write any one program using a function to print the greater of two numbers.
2. Differentiate formal and actual parameters.
3. What are built-in functions? Give three.

Model short answers

1. `int greater(int x,int y){ return (x>y)?x:y; }` call from main after `scanf`.
2. Formal: names in the definition. Actual: values in the call. They correspond in order and type.
3. Library functions: `printf`, `scanf`, `getchar`, `strlen`, `sqrt`.

Long questions (board Section C)

1. What are the main characteristics of a function? Describe with an example program.

Pointers and the address operator ★ Golden

One college period · 40–45 minutes · Pointers (Sindh XII + C paper)

What you will learn

- Explain & and *
- Declare and initialise a pointer
- Write a program that prints address and value
- Pass addresses to a function to double two numbers

Pointer. A variable that stores the memory address of another variable. Declared with *. The address operator & gives an address; the dereference operator * accesses the value at that address.

```
int x = 10;
int *p;      /* p is a pointer to int */
p = &x;      /* p holds the address of x */
printf("Value = %d\n", *p);  /* 10 */
printf("Address = %u\n", p);
```

```
void double_them(int *a, int *b) {
    *a = *a * 2;
    *b = *b * 2;
}
void main() {
    int x = 4, y = 5;
    double_them(&x, &y);
    printf("%d %d", x, y);  /* 8 10 */
}
```

HOW BIEK ASKS THIS

2025 Paper II: What is a pointer in C? How is it initialised? What is the address operator?

COMMON BOARD MISTAKE

p = &x is wrong for the first assignment if p is int. Use p = &x; then read *p. Mixing them loses marks.

اردو میں سمجھیں

پوائنٹر وہ متغیر ہے جو دوسرے متغیر کا پتہ رکھتا ہے۔ & پتہ دیتا ہے، * اس پتے پر موجود قدر دیتا ہے۔

CHECK YOURSELF

1. If `int x=5; int *p=&x;` what is `*p`?
2. Symbol for address operator?

Answers: 1. 5 · 2. &

MCQs

1. A pointer stores a:
 - a. value only

- b. memory address
- c. file name
- d. keyword

2. &x means:

- a. value of x
- b. address of x
- c. pointer declaration
- d. multiply x

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is a pointer in C? How is it initialised?
2. Explain the address operator with an example.
3. Write a program to display address and value of a variable using a pointer.

Model short answers

1. A variable that holds an address. `int *p; p = &x;` or `int *p = &x;`
2. `&x` is the address of x. `scanf` uses it; pointers store it.
3. Declare x and `p=&x;` `printf` value `*p` and address `p`.

Long questions (board Section C)

1. Explain pointers with a diagram of x and p. Write a function that receives two addresses and doubles both values.

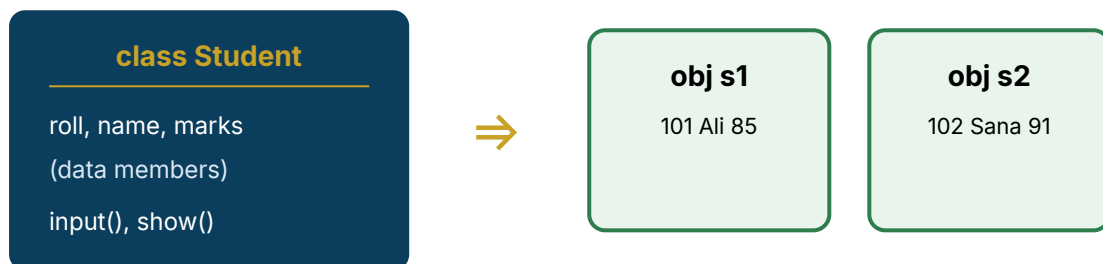
OOP in C++: class, object, constructor, inheritance, polymorphism

★ Golden

One college period · 40-45 minutes · OOP using C++ and file handling

What you will learn

- State encapsulation, inheritance, polymorphism
- Write a simple class Student with constructor
- Explain public, private, protected
- Open text/binary files for read, write, append



A class is a blueprint; objects are instances with their own data.

- Encapsulation: wrap data + functions; hide data with private.
- Inheritance: derived class reuses base class (reusability).
- Polymorphism: one name, many forms — overload (same name, different parameters) and override (same signature in derived class).

```
#include <iostream>
using namespace std;
class Student {
    int age; float percentage;           // private by default
public:
    Student() { age = 0; percentage = 0; }           // default constructor
    Student(int a, float p) { age = a; percentage = p; } // overloaded
    void show() { cout << age << " " << percentage; }
    ~Student() { } // destructor
};
int main() {
    Student s1(17, 82.5);
    s1.show();
}
```

Access specifiers: private (only inside class), public (everywhere), protected (class + derived).

Constructor has the class name, no return type, called automatically. Destructor ~ClassName() at end of life.

Inheritance sketch

```
class Base { public: int x; };
class Drive : public Base { public: int y; };
// Drive object can use x and y
```


File handling

Text files store characters; binary files store raw bytes (objects). Modes: ios::in read, ios::out write (truncate), ios::app append. BOF beginning, EOF end. Streams: get/put (char), getline (string), read/write (binary).

```
#include <fstream>
ofstream out("data.txt");
out << "Ali 80\n";
out.close();
ifstream in("data.txt");
string line; getline(in, line);
```

HOW BIEK ASKS THIS

Sindh lab: class STUDENT with AGE and PERCENTAGE; TIME class with constructors; inherit DRIVE from BASE; write/read text file; write/read student object as binary.

اردو میں سمجھیں

کلاس خاکہ ہے، آبجیکٹ اس کی کاپی۔ Encapsulation ڈیٹا چھپاتا ہے، Inheritance وراثت، Polymorphism ایک نام کئی کام۔
Constructor آبجیکٹ بنتے وقت چلتا ہے۔
فائل بینڈلنگ: ٹیکسٹ اور بائنری، read/write/append.

CHECK YOURSELF

1. Does a constructor return a value?
2. ios::app means?

Answers: 1. No — not even void · 2. Append to the file without erasing old data

MCQs

1. Wrapping data and functions together is:
 - a. inheritance
 - b. encapsulation
 - c. compilation
 - d. paging
2. A constructor has:
 - a. the class name and no return type
 - b. any name
 - c. return type int
 - d. a destructor symbol

Key: 1=b, 2=a

Short questions (3 marks each)

1. Explain class and object.
2. Differentiate constructor and destructor.
3. Differentiate text and binary files.

Model short answers

1. Class: user-defined type with data members and member functions. Object: variable of that type.
2. Constructor initialises (same name, auto-called). Destructor cleans up (~Name), no arguments.
3. Text: human-readable characters. Binary: exact memory image, efficient for objects.

Long questions (board Section C)

1. Explain OOP principles with a C++ class that uses a constructor. Discuss inheritance and polymorphism briefly.

Visual Basic IDE, forms and ActiveX controls ★ Golden

One college period · 40–45 minutes · Programming using Visual Basic

What you will learn

- Describe the VB IDE and a form
- List basic ActiveX controls and important properties
- Set Cancel button, use labels, text boxes, command buttons, combo, option, check

Visual Basic. An event-driven programming language and IDE from Microsoft. You draw controls on a Form; code runs when events occur (Click, Load, Change).

Control	Use	Key properties
Form	Window	Caption, BackColor, BorderStyle
Label	Display text the user does not type	Caption, Font, AutoSize
TextBox	Input / output of text	Text, MaxLength, PasswordChar, MultiLine
CommandButton	Click to run code	Caption, Enabled, Cancel, Default
CheckBox	Yes/No independent	Value 0/1/2
OptionButton	One choice from a group	Value True/False
ComboBox / ListBox	Pick from a list	List, Text, AddItem
Image / PictureBox	Pictures	Picture, Stretch

Cancel button property belongs to CommandButton (MCQ). Default step in For...Next is 1. DIM declares a variable. Variant can hold any type: Dim x As Variant or Dim x (undeclared = Variant if no Option Explicit — board still asks “how can a variable be defined as variant”).

```
Private Sub cmdAdd_Click()
    lblSum.Caption = Val(txtA.Text) + Val(txtB.Text)
End Sub
```

HOW BIEK ASKS THIS

2026 shorts: ComboBox properties; TextBox properties; basic ActiveX controls; Cancel button object; DIM; Option Button vs CheckBox.

اردو میں سمجھیں

Visual Basic میں فارم پر بٹن، ٹیکسٹ باکس، لیبل رکھتے ہیں۔ کلک ایونٹ پر کوڈ چلتا ہے۔ Option Button ایک انتخاب، CheckBox ہاں/نہیں۔

CHECK YOURSELF

1. Default For step?

2. Which control is for a single choice in a group?

Answers: 1. 1 · 2. Option Button (radio)

MCQs

1. The Cancel button property belongs to:
 - a. Form
 - b. Label
 - c. Button
 - d. Textbox
2. Default Step in For...Next is:
 - a. 2
 - b. 0
 - c. -1
 - d. 1
3. DIM declares a:
 - a. variable
 - b. constant only
 - c. keyword
 - d. function only
4. A control that selects a single choice from a group:
 - a. Check Box
 - b. Option Button
 - c. Caption Box
 - d. List Box

Key: 1=c, 2=d, 3=a, 4=b

Short questions (3 marks each)

1. What is a ComboBox? List properties.
2. Write properties of a TextBox.
3. What are basic ActiveX controls in VB?
4. How can a variable be defined as Variant?

Model short answers

1. Drop-down list combining text box + list. Properties: List, ListIndex, Text, Style, AddItem, Sorted.
2. Text, Font, MaxLength, PasswordChar, MultiLine, Locked, Alignment.
3. Label, TextBox, CommandButton, CheckBox, OptionButton, ComboBox, ListBox, Frame, Timer, Image.
4. Dim x As Variant or Dim x (Variant is the default type).

Long questions (board Section C)

1. Explain commonly used VB controls with properties. Write a form program that adds two numbers.

VB language: operators, if, Select Case, loops, arrays, functions

★ Golden

One college period · 40–45 minutes · Programming using Visual Basic

What you will learn

- Write If...Then and Select Case
- Write For...Next, Do While, Do Until
- Declare arrays and small functions
- Translate formulae into VB

```
If marks ≥ 40 Then
    lblR.Caption = "Pass"
Else
    lblR.Caption = "Fail"
End If
```

```
Select Case ch
    Case 1: lbl.Caption = "Add"
    Case 2: lbl.Caption = "Sub"
    Case Else: lbl.Caption = "Invalid"
End Select
```

```
Dim i As Integer, s As Integer
s = 0
For i = 0 To 50 Step 2
    s = s + i      ' even numbers 0-50
Next i
```

```
' sum and average of 10 inputted numbers
Dim i As Integer, n As Double, t As Double
t = 0
For i = 1 To 10
    n = Val(InputBox("Number " & i))
    t = t + n
Next i
MsgBox "Sum=" & t & " Avg=" & t / 10
```

Maths	VB
$x = 3ab^2 / c$	<code>x = 3 * a * b ^ 2 / c</code>
$Y = (4/3)\pi r^3$	<code>Y = (4 / 3) * 3.14 * r ^ 3</code>
$Y = (a+b)^n$	<code>Y = (a + b) ^ n</code>
$a = \frac{1}{2} \text{ base} \cdot \text{height}$	<code>a = 0.5 * base * height</code>

Built-in functions: Val, Str, Left, Right, Mid, Len, UCase, LCase, Int, Rnd, Date, Sqr, Abs, Format. User-defined: Function Area(r As Double) As Double : Area = 3.14 * r * r : End Function. Errors: syntax, run-time, logical. On Error GoTo handler.

HOW BIEK ASKS THIS

2026 programs: even 0–50; sum and average of ten numbers. Long: arrays; loops; built-in function with a program.

اردو میں سمجھیں

VB میں If...Then، Select Case، For...Next اور Do While لوپ ہیں۔ ^ طاقت ہے۔ Val ٹیکسٹ کو عدد بناتا ہے۔

CHECK YOURSELF

1. Even numbers 0–50: Step?

2. Val("12a") in VB6?

Answers: 1. 2, start 0 · 2. 12 (stops at first non-numeric, classic VB)

MCQs

- Comparison operators are also called:
 - Comparators
 - Comparables
 - Relations
 - Relational
- The last statement in a Function is often:
 - Footer
 - Header
 - End Function / Return value assignment
 - Call
- A function returns a:
 - form
 - value
 - only a MsgBox
 - table

Key: 1=d, 2=c, 3=b

Short questions (3 marks each)

- Write the structure of Select Case.
- Write a VB program to generate even numbers 0–50.
- Write a VB program to calculate sum and average of ten numbers.
- Convert $Y = (4/3)\pi r^3$ to VB.
- What is an error? Define its types.

Model short answers

- Select Case expr : Case v1: ... : Case Else: ... : End Select
- For i = 0 To 50 Step 2 : Print i : Next
- Loop 10 InputBox, add to total, divide by 10.
- $Y = (4 / 3) * 3.14 * r ^ 3$
- Mistake in a program. Syntax (grammar), run-time (divide by zero), logical (wrong formula, runs but answer wrong).

Long questions (board Section C)

1. Explain arrays in VB with types and an example. OR Explain loops with two types and programs. OR Built-in functions with a program.

DBMS, models, ER diagrams, keys and normalisation ★ Golden

One college period · 40–45 minutes · Database fundamentals

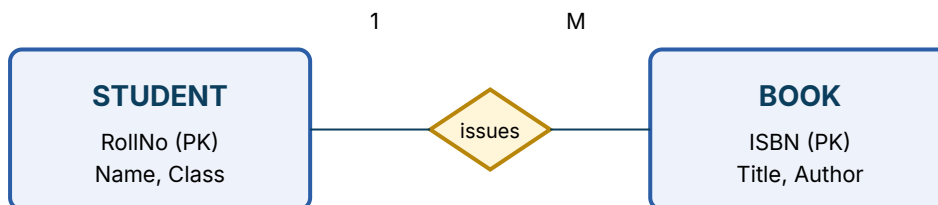
What you will learn

- Differentiate file system and DBMS
- Draw hierarchical, network, relational, object models
- Define PK, FK, candidate, super key; 1NF 2NF 3NF
- Draw a simple ER diagram (library / student-book)

Database. An organised collection of related data. DBMS is software that creates, manages and controls access to the database (MS Access, Oracle, MySQL).

Advantages vs flat files: less redundancy, data integrity, sharing, security, backup, query language. Components: hardware, software, data, procedures, people, DML/DDL (data access language). Users: DBA, application programmer, end user.

Model	Picture
Hierarchical	Tree — one parent, many children (IMS)
Network	Graph — many-to-many with pointers (CODASYL)
Relational	Tables (relations) linked by keys — SQL works here
Object / object-relational	Objects with methods; ORDBMS



Entity Relationship: rectangles = entities, diamond = relationship, ovals = attributes (in full ER).

Term	Meaning
Table / relation	A grid of rows and columns
Record / tuple	A row
Field / attribute	A column
Primary key	Unique identifier, not null
Foreign key	A field that matches the PK of another table
Candidate key	Any field(s) that could be PK
Super key	Any set that uniquely identifies, may have extras
1NF	Atomic cells, no repeating groups
2NF	1NF + non-key fields depend on whole PK
3NF	2NF + no transitive dependence on non-keys

Relationships: 1:1, 1:M, M:M, M:1. Integrity: NOT NULL, PRIMARY KEY, FOREIGN KEY (referential). Three ANSI levels: internal/physical, conceptual/logical, external/view.

HOW BIEK ASKS THIS

Both options 2026: PK vs FK; table, query, report; DBMS components; data models long question; Access data types long question.

اردو میں سمجھیں

ڈیٹا بیس منظم متعلقہ ڈیٹا ہے۔ DBMS سافٹ ویئر (Access) اسے سنبھالتا ہے۔ Primary Key یکتا، Foreign Key دوسری ٹیبل کا لنک۔

Relational ماڈل ٹیبلز اور SQL۔ 1NF 2NF 3NF ڈیٹا دہرانے سے بچاتے ہیں۔

CHECK YOURSELF

1. Row is also called?
2. SQL works on which model?

Answers: 1. Record / tuple · 2. Relational

MCQs

1. A unique key in a table is the:
 - a. Primary key
 - b. Foreign key
 - c. Super key only
 - d. Candidate only — PK is the chosen candidate
2. The primary object in a database is the:
 - a. Table
 - b. Query
 - c. Report
 - d. Form
3. Tree-like database model:

- a. Relational
- b. Hierarchical
- c. Network
- d. Object

4. Columns in a database are:

- a. Tuples
- b. Attributes
- c. Keys only
- d. Tables

5. The model that allows SQL is:

- a. Network
- b. Hierarchical
- c. Relational
- d. Flat

6. DBMS advantage:

- a. data depends on programs
- b. redundancy increases
- c. data is integrated and shared
- d. integrity decreases

Key: 1=a, 2=a, 3=b, 4=b, 5=c, 6=c

Short questions (3 marks each)

1. Differentiate primary key and foreign key.
2. Define table, query, report.
3. Name the major components of DBMS.
4. Who is a DBA? Three responsibilities.
5. Define one-to-many relationship.
6. Write any four advantages of DBMS.
7. Name four objects in MS Access.

Model short answers

1. PK uniquely identifies a row in its own table. FK is a field pointing to a PK in another table to link records.
2. Table stores data. Query asks questions / filters. Report is a printable layout of data.
3. Hardware, software, data, procedures, users, query language.
4. Database Administrator: security, backup, performance, user accounts, recovery.
5. One parent row links to many child rows (one class, many students). Draw 1 — M.
6. Less redundancy, integrity, sharing, security, standard SQL, backup.
7. Tables, Queries, Forms, Reports (Macros, Modules).

Long questions (board Section C)

1. What is a data model? Explain types with diagrams. OR Explain ER modelling and normalisation up to 3NF with a library example.

MS Access objects, data types and SQL ★ Golden

One college period · 40–45 minutes · Database fundamentals

What you will learn

- List Access objects and data types for a long answer
- Write CREATE, INSERT, UPDATE, DELETE, SELECT
- Plan a Library Management schema

Object	Purpose
Table	Store data in fields/records; set PK
Query	Select, update, calculate, join tables
Form	On-screen data entry (like a VB form)
Report	Print marksheets, lists
Macro / Module	Automate / VBA

MS Access data types (learn for Section C)

Type	Stores
Text / Short Text	Names, up to 255 chars
Memo / Long Text	Remarks, long notes
Number	Byte, Integer, Long, Single, Double
Date/Time	Dates and clocks
Currency	Money, 4 decimal
AutoNumber	Automatic PK
Yes/No	Boolean
OLE Object	Pictures, files
Hyperlink	URLs
Attachment / Lookup	Newer Access extras

```

CREATE TABLE Book (
  ISBN TEXT PRIMARY KEY,
  Title TEXT,
  Author TEXT,
  Price CURRENCY
);
INSERT INTO Book VALUES ('978-1', 'CS XI', 'STBB', 450);
SELECT Title, Author FROM Book WHERE Price > 400;
UPDATE Book SET Price = 500 WHERE ISBN = '978-1';
DELETE FROM Book WHERE ISBN = '978-1';

```

SELECT is projection (columns) + WHERE is selection (rows). JOIN combines tables on keys. Equijoin: WHERE Book.AuthorID = Author.ID. Operators: AND OR NOT, = <> > < LIKE BETWEEN.

Planning a library database

1. Purpose: issue and return books.
2. Tables: STUDENT(Roll PK, Name, Class), BOOK(ISBN PK, Title, Author), ISSUE(IssueID PK, Roll FK, ISBN FK, IssueDate, ReturnDate).
3. Unique fields: Roll, ISBN, IssueID.
4. Relationships: Student 1—M Issue M—1 Book, all up to 3NF (no repeating books inside student).
5. Refine: add Fine, status Issued/Returned.

HOW BIEK ASKS THIS

Practical 2026 Access: create Library_Management, PK, 5 records, query currently issued books.
Theory: explain all Access data types.

اردو میں سمجھیں

Access میں Table ڈیٹا، Query سوال، Form انٹری، Report پرنٹ، SQL: SELECT چننا، INSERT داخل، UPDATE بدل، DELETE مٹانا۔

CHECK YOURSELF

1. Which object is for printing?
2. AutoNumber is typically used as?

Answers: 1. Report · 2. Primary key

MCQs

1. MS Access object used to display and print data:
 - a. Report
 - b. Table
 - c. Form
 - d. Macro
2. This key uniquely identifies a record:
 - a. Super Key
 - b. Foreign Key
 - c. Conditional Key
 - d. Primary Key

Key: 1=a, 2=d

Short questions (3 marks each)

1. Write SELECT syntax with WHERE.
2. Differentiate form and report.
3. Write steps to design a library database.

Model short answers

1. SELECT field1, field2 FROM table WHERE condition; example SELECT Title FROM Book WHERE Price>400;
2. Form is interactive on screen; report is formatted for paper/PDF.
3. Purpose, tables, fields, unique keys, relationships to 3NF, refine.

Long questions (board Section C)

1. Explain all the data types used in MS-Access. OR Explain SQL DML statements with examples.

SDLC, stakeholders and phases ★ Golden

One college period · 40–45 minutes · System Development Life Cycle

What you will learn

- Define system and SDLC
- Name stakeholder roles
- Describe planning, analysis, design, coding, testing, deployment, maintenance

System. A set of related components that work together toward a goal. SDLC is the planned process of building information software from idea to retirement.

Stakeholder	Role
Project manager	Plan, cost, schedule
Business analyst	Gather requirements
System architect	Overall design
Developer	Write code
Tester	Find bugs
Operations	Deploy on servers
Production support	Maintenance after go-live

Planning → Requirement analysis → Design → Coding → Testing → Deployment →

Maintenance

- Planning: problem, feasibility (technical, economic, operational), schedule.
- Requirement engineering: functional (what it must do) vs non-functional (speed, security). SRS document.
- Design: architecture, algorithms, flowcharts, database schema, UI.
- Implementation: coding in C / VB / other.
- Testing: find bugs and errors; unit, integration, system, acceptance.
- Deployment & SLA: install, train, service-level agreement.
- Maintenance: fix, adapt, improve.

Models you may mention: Waterfall (linear, old board favourite) vs Agile (iterative sprints).

اردو میں سمجھیں

SDLC سافٹ ویئر بنانے کا منصوبہ بند چکر ہے: منصوبہ، ضرورت، ڈیزائن، کوڈ، ٹیسٹ، تنصیب، دیکھ بھال۔

CHECK YOURSELF

1. SRS means?
2. Who gathers requirements?

Answers: 1. Software Requirements Specification · 2. Business analyst

MCQs

1. SDLC is mainly about:
 - a. repairing printers
 - b. developing software in stages
 - c. only testing viruses
 - d. drawing cables
2. Converting design into programs is:
 - a. testing
 - b. coding / implementation
 - c. feasibility
 - d. retirement

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define SDLC and list its phases.
2. Differentiate functional and non-functional requirements.
3. Who are the stakeholders of SDLC?

Model short answers

1. Life cycle of software: plan, analyse, design, code, test, deploy, maintain.
2. Functional: features (print marksheet). Non-functional: quality (response in 2 seconds, password security).
3. PM, analyst, architect, developers, testers, operations, support, plus the client/users.

Long questions (board Section C)

1. Explain the System Development Life Cycle with a diagram and the role of each stakeholder.

Multimedia systems; wireless and mobile communication

One college period · 40-45 minutes · Multimedia and wireless

What you will learn

- Define multimedia components and file types
- Use layers, keyframes, export/share a short video (theory + lab idea)
- Compare Wi-Fi, Bluetooth, infrared, WiMAX, microwave
- Outline cellular network, 2G-5G, GEO/MEO/LEO, ad hoc / MANET

Multimedia. The combination of text, graphics, animation, audio and video used to communicate. Advantages: richer teaching, advertising, games, training.

Component	Typical files
Text	.txt .doc .pdf
Graphics	.jpg .png .gif .bmp .svg
Animation	.gif .mp4
Audio	.mp3 .wav .aac
Video	.mp4 .avi .mkv

Packages: web apps vs desktop (ProShow Gold, JetAudio, Shotcut, Canva, PowerPoint). Scene/canvas = work area; layers stack pictures/text; keyframes mark motion in animation. Lab: 2-minute video, export MP4, upload/share.

Wireless

- Wireless: no physical cable; uses electromagnetic waves. Mobile: communication while moving.
- Terms: frequency spectrum, radio signal, transceiver, access point, line of sight.

Short range	Typical use
Wi-Fi	LAN internet in college / home
Bluetooth	earbuds, file share, few metres
Infrared	TV remote, line of sight
Microwave	dishes, backhaul
WiMAX	city-wide wireless (MAN-like)

Satellites: GEO ~36000 km (TV), MEO (GPS), LEO (phones, Starlink-style). Cellular: cell, BTS, BSC, MSC, frequency reuse. Generations: 2G GSM voice, 3G GPRS/data, 4G LTE fast data, 5G low latency. Ad hoc: no fixed AP; MANET is a mobile ad hoc network.

اردو میں سمجھیں

ملٹی میڈیا متن، تصویر، آواز، ویڈیو، اینیمیشن کا ملاپ ہے۔ وائرلیس تار کے بغیر ریڈیو لہریں: وائی فائی، بلوٹوتھ، سیلولر 2G سے 5G۔

CHECK YOURSELF

1. Bluetooth vs Wi-Fi range?
2. 4G technology name often asked?

Answers: 1. Bluetooth shorter; Wi-Fi room/building · 2. LTE

MCQs

1. MP3 is mainly a file of:
 - a. video
 - b. audio
 - c. database
 - d. source code
2. Wi-Fi is a:
 - a. wired LAN cable
 - b. short-range wireless technology
 - c. printer type
 - d. CPU register
3. GSM is associated with:
 - a. 2G
 - b. 5G only
 - c. fibre optics
 - d. OSI layer 7 only

Key: 1=b, 2=b, 3=a

Short questions (3 marks each)

1. Define multimedia and its components.
2. Differentiate Wi-Fi and Bluetooth.
3. What is a cellular network? Name 2G to 5G.
4. GEO vs LEO satellites.

Model short answers

1. Combination of text, graphics, animation, audio, video. Used in education and ads.
2. Wi-Fi: higher speed/range for internet. Bluetooth: very short range for accessories.
3. Geographic area split into cells with BTS. 2G GSM, 3G, 4G LTE, 5G.
4. GEO high and stationary over earth (TV). LEO low and fast (phones, imaging).

Long questions (board Section C)

1. Explain multimedia components and how you would make a 2-minute educational video. OR Explain wireless technologies and cellular generations.

Paper II Option I mock (C + Access) with answers ★ Golden

One college period · 40–45 minutes · Revision

What you will learn

- Sit a full Option I paper and check against keys

Section A key (15)

1 main() 2 C (Ritchie) 3 Programmer 4 Compiler 5 5number invalid 6 % with int 7 k++ post 8 / and % highest among listed arithmetic 9 Primary key 10 continue 11 data integrated 12 Table 13 identifiers 14 Iteration 15 /* */

Section B — practise these 10

1. Skeleton of a C program.
2. #include and #define examples.
3. Program: table of n.
4. break vs continue.
5. Three reserved words.
6. Address operator example.
7. Factorial program.
8. scanf vs getchar.
9. PK vs FK.
0. Evaluate integer expressions.
1. DBMS components.
2. greater() function program.
3. Three maths→C conversions.
4. Nested pattern 1 / 12 / 123... and number-square table.
5. Table, query, report.

Section C — pick 3

1. switch with diagram.
2. Operators in C.
3. Characteristics of functions + program.
4. Data models in DBMS.
5. MS Access data types.

WORKED EXAMPLE / BOARD STYLE

Scoring a C program on the board: correct headers (1), main (1), declarations (1), logic/loop (3), output (1), indentation/comments (1). Always write clrscr/getch if your college uses Turbo C, and box the output.

اردو میں سمجھیں

آپشن I: C کے پروگرام کاغذ پر صحیح semicolon اور & کے ساتھ لکھیں۔ Access کے ڈیٹا ٹائپ زبانی یاد کریں۔

MCQs

1. Option I mock: the required function in every C program is:
 - a. main()
 - b. start()
 - c. loop()
 - d. file()

Key: 1=a

Long questions (board Section C)

1. Write the factorial program and the switch-menu program from memory in 25 minutes.

Paper II Option II mock (VB + Access) with answers ★ Golden

One college period · 40–45 minutes · Revision

What you will learn

- Sit a full Option II paper and check against keys

Section A key (15)

1 Command Button (Cancel) 2 Step default 1 3 Record 4 Report 5 Primary Key 6 Check Box is 0/1 (paper wording) / Option for exclusive — learn both 7 Relational operators 8 Pass by value 9 Return/End Function 10 Attributes 11 Hierarchical 12 Option Button 13 Function 14 Variable 15 Relational model

Section B list from 2026

1. Four advantages of DBMS.
2. ComboBox + properties.
3. Variant type.
4. One-to-many + diagram.
5. Four Access objects.
6. Select Case structure.
7. DBA responsibilities.
8. Basic ActiveX controls.
9. ADO MDI SQL DBA OLE full forms (any three).
0. TextBox properties.
1. Three VB formulae.
2. Types of errors.
3. Even numbers 0–50 program.
4. Sum and average of ten numbers.
5. PK vs FK.

Section C

1. Arrays and types with examples.
2. Loop — any two types with examples.
3. Built-in function + program.
4. All database models with diagrams.
5. All MS Access data types.

WORKED EXAMPLE / BOARD STYLE

ADO = ActiveX Data Objects. MDI = Multiple Document Interface. SQL = Structured Query Language. DBA = Database Administrator. OLE = Object Linking and Embedding.

اردو میں سمجھیں

آپشن II: فارم کے نام cmd، txt، lbl رکھیں۔ Val سے جمع کریں۔ Access کے سوال C والے کاغذ جیسے ہی ہیں۔

MCQs

1. ADO stands for:
 - a. ActiveX Data Objects
 - b. Access Data Only
 - c. All Data Output
 - d. Array Data Object

Key: 1=a

Long questions (board Section C)

1. Design on paper a VB form with two text boxes, four command buttons (+ − × ÷) and a label for the result. Write the four Click events.